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AI-Driven Identification of Reference Projects for Architectural Tenders: A Data-Driven Approach

Development of a Project Retrieval System and its Application
in the AEC Industry

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Abstract

The identification of suitable reference projects is a critical yet time-consuming aspect of the architectural tendering process. This thesis investigates how artificial intelligence (AI) can be leveraged to automate and optimize this task, focusing on Cedervall Arkitekter as a case study. A data-driven retrieval system was developed to mine internal datasets—specifically the Milltime database—encompassing both structured project metadata and unstructured user notes. After evaluating multiple AI methods, an embedding-based retrieval approach integrated with keyword filtering was selected, striking a balance between computational efficiency and retrieval accuracy.

Deployed on-premise as a web application, the final solution enables architects and procurement staff to query project records using natural language inputs. The system applies semantic similarity modeling and a customized ranking algorithm to provide rapid, relevant search results, cutting manual search time by more than 50% according to user testing. Structured interviews further demonstrated its capacity to enhance the reference project selection process and reduce reliance on personal memory. Taken together, these findings underscore the value of AI-driven retrieval systems in architectural practices, while highlighting promising directions for expanded machine learning integration within tendering and other knowledge-intensive workflows in the architecture, engineering, and construction (AEC) sector.

Key words

Artificial intelligence, Architecture, Tenders, Data Engineering, Semantic similarity, Natural Language Model, Data-driven solution

Sammanfattning

Identifieringen av lämpliga referensprojekt är en avgörande men tidskrävande del av anbudsprocessen inom arkitektur. I detta examensarbete undersöks hur artificiell intelligens (AI) kan användas för att automatisera och effektivisera denna uppgift, med Cedervall Arkitekter som fallstudie. Ett datadrivet söksystem utvecklades för att analysera interna datakällor—särskilt Milltime-databasen—som innehåller både strukturerad projektinformation och ostrukturerade användarnoteringar. Efter att flera AI-metoder utvärderats valdes en lösning baserad på inbäddade representationer i kombination med nyckelordsfiltrering, vilket ger en god balans mellan beräkningskostnad och söknoggrannhet.

Systemet, som körs lokalt som en webbapplikation, gör det möjligt för arkitekter och upphandlingspersonal att söka projektposter med hjälp av naturliga språk. Med semantisk likhetsberäkning och en anpassad rankningsalgoritm levererar systemet snabba och relevanta träffar, vilket kortar ned manuell söktid med över 50%. Strukturerade intervjuer visar dessutom att verktyget förbättrar processen för att hitta referensprojekt och minskar beroendet av personliga minnesanteckningar. Sammantaget belyser resultaten hur AI-baserade söksystem kan gynna arkitektverksamheter och öppnar för ytterligare användning av maskininlärning inom anbudsarbete och andra kunskapsintensiva områden i arkitektur-, ingenjörs- och byggsektorn (AEC).

Nyckelord

Artificiell intelligens, Arkitektur, Anbud, Dataingenjörskonst, Semantisk likhet, Naturlig språkmodell, Datadriven lösning

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Chapter 1

Introduction

The foundation of artificial intelligence (AI) can be traced back to the theoretical work of British logician Alan Turing in the mid-20th century. In 1935, Turing proposed the concept of a universal Turing machine, an abstract computing model with limitless memory that could read and write symbols based on a stored program. His ideas on heuristic problem solving emerged during World War II while he worked as a cryptanalyst at Bletchley Park, where he contemplated machines that could learn from experience. In a 1947 lecture, Turing articulated the vision of machines that could adapt their own instructions, a concept he further developed in his unpublished 1948 report, "Intelligent Machinery." Turing's pioneering insights laid the groundwork for modern AI, including the training of artificial neural networks [1].

Building on Turing's foundational concepts, the emergence of artificial intelligence can be traced back to the 1950s, marked by the groundbreaking 1956 Dartmouth Conference, which brought together key figures such as John McCarthy and Marvin Minsky. This era was characterized by the use of early digital computers, such as the EDSAC and IBM 701, which employed symbolic processing and rudimentary programming techniques to mimic intelligent behavior. Unlike today's advanced neural networks, early AI relied heavily on programmed algorithms and decision trees, which facilitated basic learning and problem-solving tasks [2].

Often recognized for his pioneering work, Arthur Samuel's checkers program, developed on the IBM 701, introduced the concept of "machine learning," allowing the program to improve by learning from past games. By employing a technique called alpha-beta pruning, it evaluated winning probabilities without analyzing every possible outcome [3]. While Samuel's work laid the groundwork for machine learning, which focuses on systems improving

through experience, artificial intelligence encompasses a broader range of capabilities, including reasoning and problem-solving.



Figure 1.1: Arthur Samuel’s checkers program on the IBM 7090 in 1959 [3]

Fast forward to the modern age, and the influence of artificial intelligence has become increasingly pervasive across various sectors. The evolution of large language models (LLMs) has been particularly significant in the last five years, demonstrating remarkable performance in diverse tasks. Prior to 2017, most natural language processing (NLP) models were limited to supervised learning for specific applications, constraining their versatility. The introduction of the Transformer architecture in 2017 marked a pivotal shift, leading to the development of groundbreaking models such as BERT and GPT. These models employ a semi-supervised approach that combines unsupervised pre-training with supervised fine-tuning, allowing for enhanced generalization capabilities [4].

Among these advancements, GPT models stand out, particularly GPT-3, which features 175 billion parameters and excels in generating contextually relevant text. As one of the most recognized LLMs, ChatGPT exemplifies the practical application of this technology, leveraging user-centered design principles to optimize its interactions [4]. Similarly, DeepSeek has recently emerged as a notable advancement in the field, offering innovative approaches to enhancing the efficiency and accuracy of AI-driven tasks. Its unique architecture and training methodologies have demonstrated significant potential in addressing complex challenges across various domains [5]. The rapid advancements in LLMs, including models like GPT-3 and DeepSeek, signal a promising future for AI, with the potential to transform various fields, including the architecture, engineering, and construction (AEC) industry [6].

1.1 Background

The AEC industry encompasses the collaborative efforts of professionals involved in the design, planning, and construction of buildings and infrastructure. This multidisciplinary field integrates architects who conceptualize designs, engineers who ensure structural integrity and functionality, and construction teams that execute the plans on-site [7].

Despite its critical role in shaping the built environment, the AEC industry is often regarded as one of the least digitized sectors globally, with many stakeholders acknowledging a longstanding resistance to change. This lack of digitization, coupled with manual processes, complicates project management, making it inefficient and tedious [8]. However, artificial intelligence is starting to reshape this landscape by enhancing operations, processes, safety, and productivity. In recent years, AI has emerged as a valuable asset in the construction market, which was valued at approximately US\$429.20 million in 2018. This technology offers significant potential to support architects in analyzing designs, expand their capabilities, and optimize decision-making through improved communication between human workers and machines. The integration of AI, therefore, represents a promising avenue for innovation in the AEC industry [6].

1.1.1 Cedervall Arkitekter

As a prominent architecture firm in Sweden, Cedervall Arkitekter is dedicated to enhancing architecture to contribute positively to society. The firm continuously develops its skills and proposes innovative solutions, focusing on the quality and sustainability of public projects such as preschools, universities, hospitals, and nursing homes. With a diverse workforce of around 100 employees from 20 nationalities, Cedervall Arkitekter emphasizes personal and professional development to ensure high-quality outcomes for its clients. The firm is certified under ISO 9001 and ISO 14001, maintaining rigorous processes for planning, implementation, and follow-up to guarantee customer satisfaction [9].

In the competitive landscape of the AEC industry, the need for innovative technologies has become increasingly critical. Cedervall Arkitekter recognizes that improving its tender work processes—also known as procurement—is essential for securing public sector projects. This process involves preparing and submitting comprehensive bids that outline proposed solutions, project costs, and timelines, requiring significant documentation and resource

allocation. By integrating data-driven tools to automate these processes, the firm aims to reduce the time and effort involved while increasing overall efficiency, thus reinforcing its commitment to innovation and quality. As an organization well-positioned to leverage AI innovations, Cedervall Arkitekter's extensive involvement in public sector tenders reflects a diverse portfolio of projects [10].

A significant challenge in this competitive environment is the identification of similar past projects that can serve as references, which is crucial for crafting compelling proposals. By exploring data-driven solutions to streamline this identification process, the firm seeks to enhance both the speed and accuracy of its tender preparations.

1.2 Problem

The AEC industry faces several challenges that hinder efficiency and innovation. One significant area of concern is the tendering process, which remains complex and resource-intensive. This section discusses the specific problems identified at Cedervall Arkitekter and formulates the scientific and engineering questions that this thesis aims to address.

1.2.1 Original Problem

As previously discussed, the AEC industry has historically demonstrated a resistance to change, and its tendering processes are no exception. Tendering involves a complex and multifaceted approach, where no single solution applies universally; strategies must be tailored to identify the most efficient methods for each project. Consequently, there are numerous aspects of the tendering process that present opportunities for improvement [11].

At Cedervall Arkitekter, a significant bottleneck has been identified in their tendering process. When local governments issue calls for tenders, firms like Cedervall participate by submitting detailed bids, a process that will be explained in more detail in Section 2.1. These bids outline the firm's objectives, the resources allocated (including personnel and their expertise) and provide a comprehensive cost estimate and implementation descriptions. A critical component of this process involves identifying relevant past projects that can serve as references to showcase the firm's capabilities and support the proposed estimates. However, with thousands of completed projects, locating the most applicable reference projects is labor-intensive and characterized by

considerable uncertainty, particularly since there is no guarantee that the most relevant projects have been identified.

This challenge is compounded by the need for extensive documentation and numerous meetings to create the bids, making the entire process both resource-heavy and time-consuming. Moreover, the effort invested can often feel unrewarding if the tender is awarded to another firm, leading to a sense of wasted resources. This highlights a clear opportunity for automation within their internal processes.

1.2.2 Scientific and Engineering Problems

In response to the growing significance of artificial intelligence in the AEC industry, Cedervall Arkitekter aims to explore how AI can enhance its operations, particularly in procurement processes. This investigation leads to a broader question regarding the potential of AI to optimize processes within architecture firms, especially in overcoming challenges associated with utilizing internal unstructured data.

Therefore, the **scientific problem** in this thesis is:

How can artificial intelligence techniques be leveraged to optimize processes across the AEC industry, with a focus on improving reference project identification for tender preparation as a case study?

The **engineering problem** in this thesis is:

Developing a solution that automates the identification of reference projects to enhance the efficiency of the tendering process at Cedervall Arkitekter.

This inquiry not only addresses Cedervall Arkitekter's immediate challenges but also seeks to contribute valuable insights and methodologies for the broader engineering and AI communities.

1.3 Purpose

The primary purpose of this thesis is to investigate how AI-driven methods can enhance the efficiency of tendering processes within architecture firms by automating the identification of suitable reference projects from unstructured internal data. Cedervall Arkitekter, with its extensive project database, serves as the focal case study. By developing and testing a prototype system, the

thesis aims to streamline the retrieval of project references, thereby improving bid accuracy, saving time, and strengthening the firm's competitiveness in securing public sector tenders.

In addition to addressing practical challenges at Cedervall Arkitekter, this research seeks to advance academic and professional discourse on AI applications within the AEC industry. By exploring techniques for processing large volumes of textual data, the thesis contributes to existing knowledge on optimizing internal workflows in architectural practices. Insights gained and methodologies developed may serve as a guide for other organizations facing similar hurdles, fostering further research and innovation in AI-driven solutions across the architectural and engineering landscape.

1.3.1 Ethical and Sustainability Considerations

The integration of AI-driven solutions in the tendering process at Cedervall Arkitekter not only addresses operational inefficiencies but also aligns with broader ethical and sustainability goals, particularly the United Nations Sustainable Development Goals (SDGs) 8 and 9.

Goal 8: Decent Work and Economic Growth

SDG 8 emphasizes the promotion of sustained economic growth and decent work for all [12]. By automating repetitive tasks within the tendering process, Cedervall Arkitekter enables its skilled workforce to focus on more meaningful work, enhancing productivity and job satisfaction. This shift contributes to the overall economic performance of the firm, enabling higher-quality outcomes for clients.

Goal 9: Industry, Innovation, and Infrastructure

SDG 9 focuses on building resilient infrastructure and fostering innovation [13]. The adoption of AI can support Cedervall Arkitekter in its commitment to developing sustainable community properties. Automation allows the firm to allocate more resources toward innovative and environmentally responsible architectural solutions.

1.4 Goals

Building on the overarching purpose of enhancing tendering efficiency through AI-driven methods, this chapter delineates the primary goal of the project—developing

a tailored solution at Cedervall Arkitekter—and introduces three specific sub-goals guiding its design, integration, and evaluation.

Sub-goal 1: System Development

Develop a proof-of-concept data-driven system that automates the identification of relevant reference projects from Cedervall Arkitekter’s extensive project database. This system uses data analysis techniques, such as semantic similarity, to evaluate past projects and recommend the most relevant references for new bids.

Sub-goal 2: User Integration

Ensure the developed system is user-friendly and integrates seamlessly into Cedervall Arkitekter’s existing workflows. This includes gathering user feedback during the development process to refine the interface and functionality according to the needs of the firm’s employees.

Sub-goal 3: Evaluation and Documentation

Assess the effectiveness of the AI-driven system in improving the tendering process through performance metrics such as time savings and accuracy of reference project identification. Document the findings and provide recommendations for further enhancements and broader applications within the AEC industry.

Achieving these goals is expected to streamline Cedervall Arkitekter’s tendering process, making it more efficient and responsive to public sector opportunities. Beyond immediate practical benefits, the project will also yield insights and methodologies that can inform future AI-driven innovations across the AEC industry.

1.5 Structure of the Thesis

This thesis is structured into eight main chapters, each addressing a critical aspect of the research, development, and evaluation of the AI-driven reference project retrieval system for Cedervall Arkitekter.

Introduction provides an overview of the thesis, outlining the background and motivation for the research. It introduces the challenges faced by architecture firms in identifying reference projects, formulates the research problem,

and presents the purpose and goals of the study. The chapter also includes ethical and sustainability considerations and provides an outline of the thesis structure.

Background explores key concepts relevant to the thesis. It covers procurement processes, with a particular focus on public procurement in Sweden, and examines how AI is applied in the AEC industry. Additionally, it reviews existing AI-enabled procurement solutions, highlighting their functionalities and limitations.

Data describes the data sources used in this research, including Cedervall Arkitekter's internal Milltime database. It details the structured and unstructured data utilized for project retrieval, the process of acquiring and preprocessing the data, the database structure, and security considerations.

Methods presents the technical methodologies employed in developing the AI-driven solution. It discusses the selection of natural language processing (NLP) models, the semantic similarity approach, and the ranking algorithm used to retrieve and prioritize relevant reference projects. Additionally, it covers the system's computational infrastructure, user interface development, and experimental design.

Development details the practical implementation of the system. It describes the iterative development process, from initial problem analysis and data exploration to the design and deployment of the final tool. The chapter also covers key decisions made regarding the transition from cloud-based models to an on-premise solution and the integration of user notes for improved project identification.

Results presents the outcomes of the study, including an overview of the final product and user interface. It summarizes the findings from structured interviews with Cedervall Arkitekter's procurement team, evaluating the system's impact on efficiency and accuracy. Broader implications for AI-driven procurement solutions in the AEC industry are also discussed.

Discussion critically evaluates the findings, addressing both the scientific and engineering challenges encountered during the project. It highlights the system's strengths, potential limitations, and areas for improvement. Furthermore, it outlines opportunities for future research and development.

Conclusion provides a summary of the thesis, emphasizing the contributions made and the practical impact of the developed solution. It reflects on the broader significance of AI-driven retrieval systems in the AEC industry and suggests potential directions for future enhancements.

Chapter 2

Background

This chapter provides an overview of procurement processes, with a particular focus on public procurement in Sweden. It examines the applications of AI in the AEC industry, emphasizing how AI can be utilized in procurement. The chapter also reviews existing AI-enabled procurement solutions on the market, highlighting their relevance to this project and the differences between them. Finally, it summarizes the key points discussed and sets the stage for how these insights will inform the project's development.

2.1 Procurement

Procurement, often referred to as tendering, is the structured process of acquiring goods, services, or works from external suppliers. This process is crucial for organizations in both the public and private sectors, ensuring the efficient and transparent acquisition of resources necessary to fulfill operational needs. A successful procurement process encompasses several key steps, including identifying needs, preparing tender documents, inviting bids, evaluating submissions, awarding contracts, and implementing the contract. Throughout this process, it is essential to uphold principles of fairness, transparency, and competitiveness.

2.1.1 Public Procurement: A Swedish Perspective

The public procurement process in Sweden is shaped by reforms aimed at marketization and new public management, which prioritize efficiency, transparency, and accountability. Local governments often restructure their procurement processes to centralize expertise, especially in complex sectors like

construction. This includes establishing dedicated procurement units and aligning organizational structures to manage procurement more effectively [14].

Procurement increasingly aligns with broader political goals, such as promoting social inclusion and environmental sustainability. Contracts may incorporate requirements that reflect ecological standards and social responsibility, demonstrating a commitment to achieving socio-political objectives beyond mere economic efficiency [15].

Understanding the complexity of this process is essential for firms aiming to navigate the tendering landscape effectively and leverage AI-driven solutions to improve efficiency and outcomes.

2.1.2 Steps in the Procurement Process

According to Sweden’s public procurement authority, *Upphandlingsmyndigheten*, the procurement process typically follows a sequence of key steps, as illustrated in Figure 2.1.

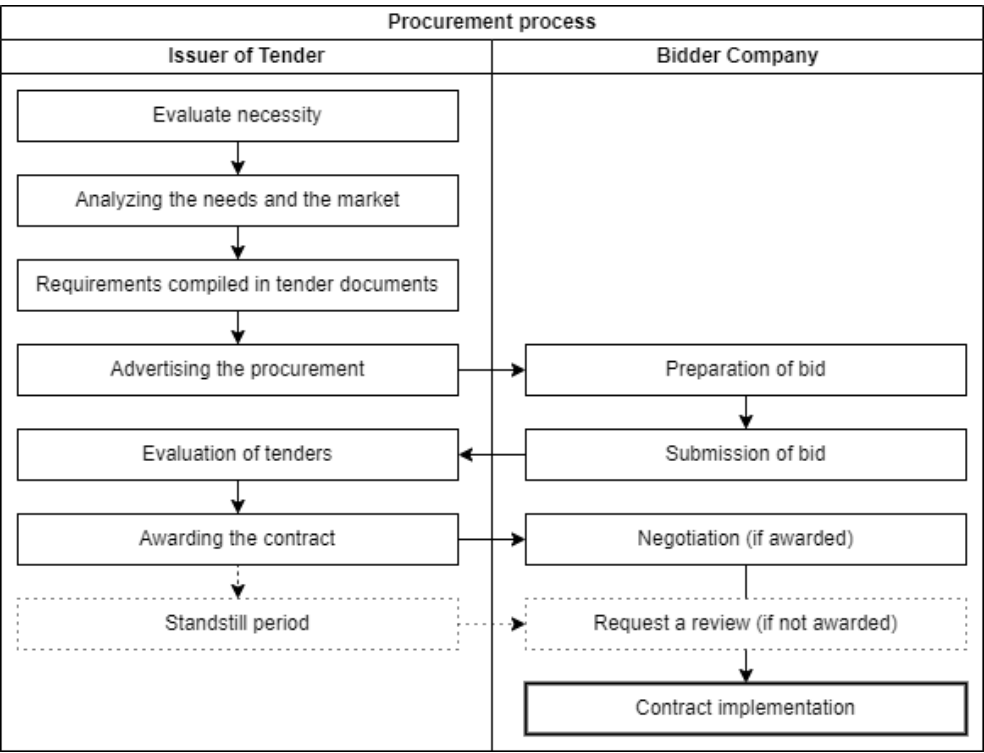


Figure 2.1: Standard procurement process in Sweden [16]

The process begins with the public organization **evaluating the necessity** of the purchase, determining whether there is a need that cannot be met internally. Once confirmed, the organization proceeds to **analyze the needs and the market**, assessing specific requirements and exploring available solutions, potentially consulting with suppliers or experts while ensuring equal treatment.

Following the analysis, the organization compiles the **requirements into tender documents**, outlining both mandatory and non-mandatory criteria to guide potential bidders.

The procurement is then **advertised** in a registered advertising database, and if the value exceeds certain thresholds, also in the EU's Tenders Electronic Daily (TED), ensuring transparency and broad access.

Interested suppliers prepare and **submit their bids**, adhering to the guidelines and deadlines provided.

After the submission deadline, the procuring organization undertakes the **evaluation of tenders**, assessing each bid against the predefined criteria to determine the offer that provides the best value and meets all mandatory requirements.

Upon completing the evaluation, the organization proceeds with the **awarding of the contract**, formally notifying the successful bidder and providing feedback to the unsuccessful ones. A **standstill period** may follow, during which unsuccessful bidders have the right to **request a review** if they believe the process was flawed.

If no challenges are raised, or once any issues are resolved, the successful bidder may engage in **negotiations** to finalize contract details, if applicable. This may involve discussing specific terms or delivery schedules.

Finally, the supplier moves on to **contract implementation**, delivering the goods or services as stipulated. The procuring organization monitors performance to ensure compliance with contractual obligations and quality standards [16].

This thesis aims to optimize the bid preparation process for bidders by offering a solution that accelerates the identification of relevant reference projects, a crucial aspect of preparing a bid.

2.2 Applications of AI in the AEC Industry

AI has increasingly been integrated into the AEC industry to address complex challenges and improve efficiency. AI methods such as genetic algorithms

(GAs), neural networks (NNs), fuzzy logic (FL), and machine learning (ML) have been widely applied, leading to significant benefits [17].

One primary application is in optimization problems. Genetic algorithms have been effectively used for project scheduling and cost optimization, assisting in finding optimal solutions for resource allocation. Neural networks and machine learning algorithms enable the analysis of large datasets for pattern recognition and prediction, aiding decision-making processes related to material performance, risk assessment, and structural analysis.

Fuzzy logic has been employed to handle uncertainty and imprecision in construction projects. By allowing for partial membership in sets, these methods provide a flexible approach to modeling real-world scenarios, enhancing decision-making under uncertain conditions [17].

Automation and robotics represent another significant application of AI in the AEC industry. Automated systems have been developed for tasks such as bricklaying and steel beam assembly, improving productivity and safety by performing repetitive or hazardous tasks [18]. Drones and autonomous vehicles are utilized for site inspections and monitoring, providing real-time data collection and analysis.

AI contributes to improved health and safety by predicting potential hazards and automating dangerous tasks, thus reducing the risk of accidents on construction sites. Additionally, AI enhances collaboration among project stakeholders through real-time data sharing, leading to better workflow management [18].

In terms of procurement processes, AI optimizes supplier selection by analyzing historical data to assess performance and reliability. Machine learning models can forecast material requirements based on project schedules, aiding in timely procurement and inventory management. AI also automates routine tasks like order processing, reducing manual effort and errors [19].

2.3 Applications of AI in Procurement

The evolving role of AI in procurement is transforming traditional workflows, introducing automation and data-driven insights to improve decision-making. While still in its early stages, AI adoption presents opportunities to enhance operational efficiency and strategic sourcing [19,20]. Procurement plays a central role in firm profitability, as companies invest substantial portions of their revenue in goods and services from suppliers. AI technologies provide practical solutions to streamline procurement activities, automate routine tasks, and support decision-making.

One key application of AI in procurement is in **semantic search and information extraction**. Natural Language Processing (NLP) techniques process both structured and unstructured data to facilitate decision-making [20]. Systems employing AI-based tools such as semantic search engines and open information extraction help professionals extract valuable information from large volumes of tender documents, enhancing transparency and efficiency. This method is particularly relevant to this project, which focuses on using AI to utilize relevant data from databases.

AI is also employed in **spend analysis and classification**. Algorithms can automatically consolidate and cleanse spending data from various sources, enabling accurate spend analysis and identifying cost optimization opportunities [19]. This leads to better budgeting, improved negotiation leverage, strategic sourcing decisions, and more accurate cost estimates for procurement bids.

2.4 Solutions on the Market

Several AI-enabled procurement platforms are currently available, demonstrating how advanced technologies can streamline processes such as spend analysis, supplier management, and decision support. Notable examples include products offered by Oracle, SAP Ariba, and GEP Smart. While these platforms share certain AI-driven functionalities, each employs unique strategies that can inform the development of custom solutions for architectural tendering.

2.4.1 Oracle Procurement Cloud

Oracle Procurement Cloud integrates artificial intelligence to facilitate data retrieval and analysis throughout procurement workflows [21]. Its analytics engine helps decision-makers compare performance metrics, cost structures, and risk factors, enabling more informed supplier assessments. By supporting automated data categorization, the platform generates actionable insights that optimize spending and supplier selection.

A notable strength of Oracle Procurement Cloud is its focus on user-centric design. AI-driven search features enable quick access to relevant documents and contract details. Conversational interfaces, such as the Oracle Digital Assistant, further streamline routine tasks like purchase order creation and supplier inquiries [22]. Although natural language queries are interpreted at a basic level, the emphasis remains on facilitating straightforward user interaction rather than sophisticated linguistic processing.

From the perspective of automating reference project identification, Oracle's approach illustrates how artificial intelligence can reduce manual data handling and enhance user experience. The system demonstrates that improving procurement workflows often requires a combination of automated analysis and intuitive user interfaces.

2.4.2 SAP Ariba

SAP Ariba focuses on integrating AI and machine learning for spend analytics and risk management, primarily by gathering procurement data on a global scale [23]. This approach makes it possible to identify cost-saving opportunities and potential threats across various categories of suppliers. It also provides real-time dashboards for strategic sourcing decisions and contract negotiations, helping users keep track of ongoing procurement activities.

In addition, SAP Ariba promotes collaboration between buyers and suppliers through its network-based approach. Chatbot functionalities manage routine tasks and retrieve information on invoices, orders, and supplier statuses, but they do not include extensive translation of natural language inputs into database queries. Instead, the platform concentrates on predictive analytics and automated alerts to enhance procurement efficiency. Such capabilities are indicative of the advantages that data-driven insights can bring to processes like reference project identification.

2.4.3 GEP Smart

GEP Smart is a procurement platform that employs AI for spend analysis, category management, and supplier performance monitoring [24]. Its cognitive computing capabilities support automated data categorization, which simplifies decision-making and helps uncover opportunities for cost optimization. Users can benefit from conversational queries that retrieve relevant procurement information, although this functionality remains centered on assisting with routine interactions rather than advanced natural language processing.

While GEP Smart does not translate user questions directly into complex database queries, its approach underlines the value of intelligently categorizing large datasets and offering guided user workflows. Such strategies are highly relevant to architecture firms aiming to automate the identification of past projects during tender preparation, since they demonstrate how AI can be used effectively to filter and organize extensive internal data.

2.5 Summary

This chapter has examined three prominent AI-enabled procurement solutions: Oracle Procurement Cloud, SAP Ariba, and GEP Smart. Each platform illustrates different methods of employing artificial intelligence for supplier assessment, cost analysis, risk management, and user-centric interfaces. Although they prioritize procurement tasks, their shared reliance on data-driven insights and automated processes offers valuable lessons for any domain dealing with large datasets.

For an architectural firm that aims to streamline tendering by identifying relevant reference projects, these examples underscore the importance of integrating automated analytics with intuitive tools for user interaction. While sophisticated natural language processing may not be essential, the ability to handle unstructured data and present findings efficiently can significantly reduce manual effort. As such, the strategies showcased by these platforms can inspire further innovation in AI-driven solutions for the AEC industry.

Chapter 3

Data

The primary data source for this project is the **Milltime** system provided by **Milient**, a Swedish company specializing in project management and time tracking solutions for professional service firms. Milient offers cloud-based tools that enable organizations to manage projects, track time and expenses, and streamline administrative processes [25].

Most of the data will be accessed via Milient's SOAP API (Simple Object Access Protocol Application Programming Interface), which allows for programmatic retrieval of information stored within the Milltime system. The project notes, however, will be retrieved through the company's internal Open Database Connectivity (ODBC) server, as the API does not provide access to this data.

3.1 Data Description

As mentioned previously, the program retrieves data from two distinct sources: Milient's SOAP API and the company's internal ODBC server, which also derives its data from the Milltime system. Table 3.1 presents the fields obtained from the SOAP API (hereafter referred to simply as the project's 'attributes'), while Table 3.2 lists the fields retrieved via the ODBC server.

Field	Description
ProjectId	Unique identifier for the project
ProjectNr	Project number
ProjectName	Name of the project
LeaderId	Identifier for the project leader
ManagerId	Identifier for the project manager
CreatedDate	Date the project was created
LastChangesDate	Date of the last changes made
Discipline	Area of expertise involved in the project
Team	Team assigned to the project
Size	Size of the project
Location	Geographic location of the project
ProjectType	Type of project (e.g., school, library)
PhaseStage	Current phase of the project
NewBuildRebuild	Indicates if it is a new build or a rebuild
PriceRange	Estimated price range of the project

Table 3.1: Project information (attributes) from the SOAP API

While this structured data offers essential high-level information, it alone is insufficient for accurately identifying relevant reference projects. To gain deeper insights, unstructured data in the form of **user notes or comments** will also be utilized.

Field	Description
ProjectId	Unique identifier for the project
RegDay	Date when the note was recorded
ActivityType	Category of the activity (e.g., internal, external)
UserNote	Detailed note entered by the employee
FullName	Full name of the employee who authored the note

Table 3.2: User note data from the ODBC server

These user notes, recorded for each hour billed on a project, provide detailed descriptions of the tasks performed, challenges faced, and specific aspects of the work. This unstructured data is crucial for understanding the unique nature and scope of each project, thereby enabling more precise identification of relevant reference projects.

3.2 Data Acquisition and Security Considerations

Access to the Milltime system's data via the SOAP API requires proper authentication and authorization to ensure data security and privacy. All interactions with the API follow Milient's data access policies and comply with relevant data protection regulations, including the General Data Protection Regulation (GDPR) [26].

In addition to SOAP-based project data, user notes are retrieved through an ODBC connection to a separate PostgreSQL database. This process also requires secure credentials and strictly adheres to internal access controls. Because these user notes may contain personally identifiable information (PII), storage and handling of the data take place on the protected internal network, which is accessible on-site via the company network and externally through a company VPN. This setup helps safeguard both project data and sensitive information throughout the entire data acquisition workflow.

3.3 Data Preprocessing

The attributes (e.g., `ProjectName`, `Location`, `LeaderId`) do not require extensive preprocessing, but the unstructured user notes demand more significant cleaning and filtering. The raw notes contain a range of irrelevant information which offer limited value for procurement purposes. To address this, filtering, cleaning, and data-merging processes are applied before creating the vector embeddings.

The program selectively retrieves user notes from the `project_reg_report` table in the PostgreSQL database via ODBC, using a filtering criteria. It only accepts usernotes marked as "external" since internal entries (e.g., administrative tasks, sick days) contain irrelevant data. It also only considers notes younger than 10 years, ensuring a focus on more current project practices. Additionally, only non-empty notes are included, discarding blank or null entries. Once the relevant rows are retrieved, each `usernote` undergoes a cleaning pipeline which includes:

- **Cleaning and Normalization:** Removal of non-printable characters and extraneous symbols to ensure consistency.
- **Lowercasing:** Converting all text to lowercase for uniform handling of tokens.

- **Stopword Filtering:** Eliminating common Swedish (or English) stop-words to focus on content-bearing words.
- **Basic Tokenization:** Splitting the text into words or phrases, which facilitates embedding generation.

By standardizing and de-noising the notes, these steps help the Sentence-BERT model focus on meaningful concepts that distinguish one project from another. This method is discussed in Section 4.2.

3.4 Database Structure

Figure 3.1 depicts the overall database schema, illustrating how the structured SOAP-derived data and the unstructured user notes are combined. While project records are stored in a local SQLite database, the user notes reside in a PostgreSQL database accessed via ODBC. The `users` table in PostgreSQL supplies names for project leaders, ensuring that only an ID is not shown on the interface.

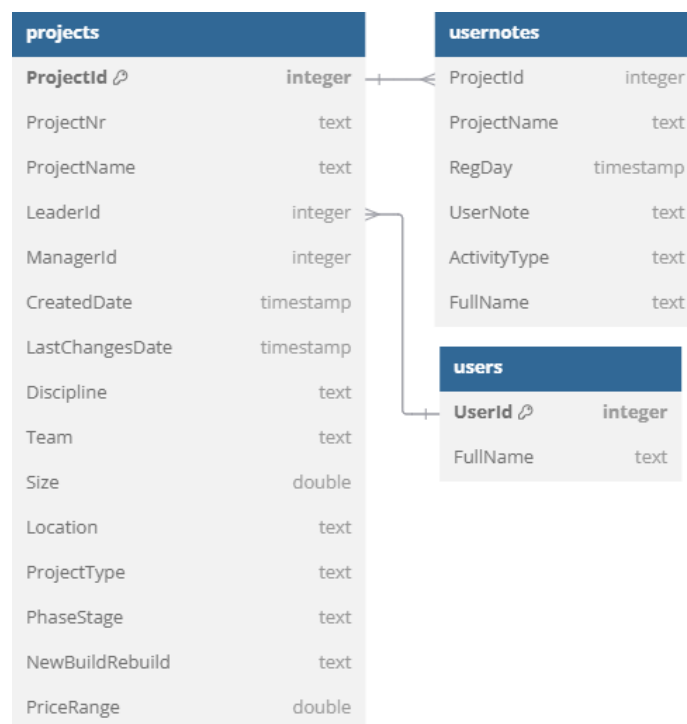


Figure 3.1: Database structure displaying how the `projects`, `usernotes`, and `users` tables relate to each other.

In the `projects` table, each record is identified by a `ProjectId`, which matches the `ProjectId` field in the `usernotes` table. This one-to-many relationship allows multiple user notes to be linked to a single project. Since the `projects` table only contains `LeaderId`, the system fetches corresponding names from the `users` table to display the project leader's `FullName` in the interface. Linking these three tables consolidates structured fields with the unstructured text of user notes, creating a unified data source for further processing and retrieval.

3.5 Storage and Data Pipeline

After merging, the final dataset is maintained in a mixed storage setup. The structured project data, fetched from Milient's SOAP API, is stored in the local SQLite database (`milltime_new.db`), while unstructured user notes, user metadata, and embeddings are kept in CSV and NumPy files. Thanks to the shared `ProjectId` key, these sources can be cross-referenced as needed.

This approach is sufficient for Cedervall Arkitekter's needs because, despite having a few hundred thousand user notes, the weekly trimming of entries older than ten years ensures that data volume remains manageable. The system does not require the concurrency handling or horizontal scaling provided by larger database solutions, as most use cases involve limited write operations and moderate search volumes. Moreover, SQLite and file-based storage reduce overhead and complexity, allowing for simpler maintenance and faster prototyping compared to full-fledged database clusters.

The weekly data pipeline is summarized in Figure 3.2. Each Sunday at 3 AM, `run-all.py` triggers the retrieval of new and updated project metadata from Milient's SOAP API and user notes from the ODBC server. During this process, outdated records exceeding the ten-year window are removed, and newly updated notes and project details are merged into the system. Afterward, the entries undergo cleaning and preprocessing before being transformed into vector embeddings. The details of this process are provided in Section 4.2. Finally, the Flask application is restarted to ensure changes are immediately reflected in any subsequent searches, keeping the data current while avoiding the overhead of a more scalable database infrastructure.

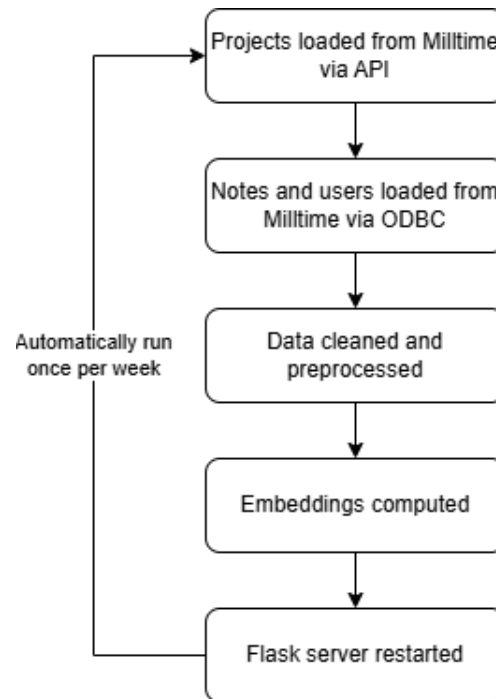


Figure 3.2: Data pipeline for weekly retrieval, cleaning, embedding computation, and Flask server restart.

Chapter 4

Methods

This project aims to develop an AI-driven solution that fetches relevant reference projects for Cedervall Arkitekter based on user-submitted queries, largely written in natural language. The overall approach was shaped by an initial investigation into state-of-the-art solutions for semantic retrieval and text analytics, followed by iterative development. Feedback loops from Cedervall Arkitekter (for practical needs) and KTH (for research methodology) informed each stage. The subsections below describe the major components considered, the final choices made, and the rationale behind those decisions.

4.1 Methodology: Approach to Method Selection

Before examining specific technologies, it is helpful to outline how the methods were chosen. The research began with a broad review of academic literature (IEEE, arXiv) and industry tools, concentrating on three key aims: effectively handling unstructured Swedish-language data, managing resource limitations, and achieving reliable accuracy. These objectives shaped a methodological framework that guided both the research phase and subsequent development.

Candidate models were then assessed for accuracy, computational requirements, multilingual capabilities, and ease of deployment on Cedervall Arkitekter's internal server. To ensure both practical relevance and academic rigor, the findings were discussed with stakeholders at Cedervall Arkitekter and with the thesis supervisor. This collaborative feedback loop led to decisions that balanced real-world constraints with the project's scientific goals, ultimately

informing the design and implementation of the chosen methods.

4.2 Semantic Similarity and Natural Language Processing

The system relies on semantic similarity techniques to compare user queries with descriptions of past projects. Initially, advanced transformer-based language models like BERT [27] and RoBERTa [28] were considered. These models are known for their strong contextual understanding and nuanced handling of language, making them attractive for text similarity tasks.

However, resource considerations and the need for multilingual processing (Swedish queries, English queries, or data needing translation) prompted a pivot toward a smaller but still robust model: a multilingual Sentence-BERT variant (`distiluse-base-multilingual-cased-v2`) [29]. This solution sufficiently captured semantic meaning across Swedish and English text while keeping memory and computational costs manageable. By encoding both the user query and the project notes into a vector space, the system computes a cosine similarity score to determine the relevance of each project. This method avoids manually defined rules and leverages learned embeddings to handle linguistic nuances more effectively.

When the user query is in English, it is first translated into Swedish using the `Helsinki-NLP/opus-mt-en-sv` model [30,31] before the similarity comparison is performed. Both the (originally Swedish or translated-to-Swedish) user query and the stored project notes are encoded into a shared vector space where relevance is judged by their cosine similarity. Specifically, for two embedding vectors $\mathbf{u}$ and $\mathbf{v}$, the cosine similarity is defined as:

$$\text{cosine_similarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

This approach avoids reliance on manually defined rules, instead leveraging learned embeddings to capture linguistic nuances such as synonyms, paraphrases, and cross-lingual semantics.

4.3 Natural Language to SQL Conversion

The original plan involved accepting natural language queries from users and translating them directly into SQL commands using specialized AI models, such as SQLCoder [32] or OpenAI’s 4o model via its API [33]. Likewise, the

Llama family of models [34] was explored for potential fine-tuning to handle domain-specific SQL generation.

After extensive testing, the `llama-3-sqlcoder-8b` model [35] proved to be the most suitable, as it consistently delivered accurate results more efficiently than other models. However, like all resource-intensive models, it required substantial computational power to operate effectively. To lessen the load, it was loaded in 4-bit precision using BitsandBytes [36], reducing both size and complexity. Although these optimizations improved performance, the overall system still proved difficult to run efficiently in a production setting.

In practice, maintaining such large models outweighed the potential benefits. After trialing smaller prototypes, it became apparent that simpler methods such as embedding-based retrieval combined with selective keyword matching for specific database fields could achieve comparable accuracy without the operational overhead. Consequently, direct end-to-end natural language-to-SQL conversion was removed from the final system design. Instead, user queries are partially parsed for keywords and translated to Swedish if needed, while a combination of manually crafted SQL statements and similarity ranked results handles data retrieval. This approach preserves the convenience of free-form queries while significantly reducing complexity and cost.

4.4 Computational Resources and Infrastructure

During the early design phase, the project explored distributed and serverless solutions, specifically Modal [37], to support large-scale, compute-intensive models. The goal was to leverage on-demand hardware scaling, avoiding the need for significant upfront investments in local infrastructure. While deploying on Modal reduced inference times and enabled resource distribution, maintaining the program round-the-clock proved costly due to the considerable consumption of credits. As a result, this approach was deemed financially impractical, prompting the exploration of alternative solutions.

With the adoption of a more lightweight Sentence-BERT-based approach, running inference on an internal server at Cedervall Arkitekter became feasible. This shift simplified data governance, reduced latency, and ensured compliance with security requirements by keeping all data within the organization's network. Consequently, the final infrastructure transitioned to on-premise deployments, eliminating reliance on cloud platforms.

4.5 User Interface Development

An intuitive user interface is essential for enabling architects and other staff to efficiently access relevant project references. The primary objective of the UI design is to accommodate general users with no programming expertise, allowing them to utilize the interface seamlessly without prior training. The interface is designed to interpret a diverse range of queries, accurately infer the user's intent, and prioritize projects that align most closely with the specified criteria. A significant aspect of the project focused on determining how to effectively achieve this goal by interpreting implicit information and identifying the key features that make a project most relevant.

Early in the planning stage, chatbot frameworks such as Rasa [38] or Botpress [39] were considered, especially for a conversational approach. However, organizational feedback indicated that a straightforward web-based form would suffice and reduce maintenance complexity.

Consequently, the final system is built using a lightweight Python Flask application. Users enter a plain-text query in a simple form, and upon submission, the application runs similarity scoring on the underlying embeddings. Project results are then ranked using a customized formula (see 4.6) and displayed in the browser. This design caters to Cedervall Arkitekter's user experience requirements, while significantly reducing the development overhead associated with dedicated chatbot frameworks. Figure 6.1 in section 6.1 presents the final interface.

4.6 Ranking

A robust ranking algorithm was crucial to effectively prioritize the most relevant projects. Multiple text retrieval techniques, such as TF-IDF [40] and BM25 [41], were tested during development but did not meet the client's precision requirements. As a result, a custom algorithm was developed to better align with their specific needs. In collaboration with the company, it was decided that the algorithm should reward projects whose attributes match the user's keywords, account for those keywords in user notes, and incorporate semantic similarity. Consequently, a three-component formula was introduced, with each component contributing to the final score.

4.6.1 Attribute Column Matches

In this step, the algorithm checks a predefined set of data columns for the presence of the user’s keywords (e.g., *ProjectName*, *Leader_fullname*, *Location*, *PhaseStage*). Each occurrence of a keyword contributes a uniform weight of $C_{\text{Attributes}}$. Points are awarded for multiple keyword matches within the same column. Formally, for each project we compute:

$$\sum_{c \in \{\text{attribute_columns}\}} \sum_{k \in \{\text{all_keywords}\}} (C_{\text{Attributes}} \times \text{countOcc}(k, \text{project}[c])) \quad (4.1)$$

where:

- $C_{\text{Attributes}}$ is the constant weight (e.g. 15) for each occurrence of a keyword in an attribute column.
- $\text{countOcc}(k, \text{project}[c])$ returns how many times keyword k appears as a substring in the project’s column c .

Alternatively, this formula can be expressed as:

$$\sum (\text{Attribute matches}) \times C_{\text{Attributes}}.$$

Example: Suppose $C_{\text{Attributes}} = 15$ and the query is “Energy-efficient school renovation in Stockholm” and the project’s name is “Södermalm International School,” with the location listed as “Stockholm” and project type listed as “School.” The keyword “Stockholm” appears therefore once and “school” twice across the attributes, giving the formula:

$$3 \times C_{\text{Attributes}} = 3 \times 15 = 45 \text{ points.}$$

4.6.2 User Note Keyword Matches

Next, the algorithm examines the user notes within each project, awarding C_{Notes} points once per matched keyword. Same number of points are given for a keyword regardless whether it appears once or many times in the notes. Mathematically:

$$\sum_{k \in \{\text{matched_keywords}\}} C_{\text{Notes}} \quad (4.2)$$

where:

- $\{\text{matched_keywords}\}$ is the set of distinct keywords found across the user's notes for that project.
- C_{Notes} is a constant factor (e.g., 10) awarded for each distinct keyword that appears in the notes.

Example: For the same query, assume $C_{\text{Notes}} = 10$ and the project has user notes containing:

- *"Renovation plans for the school focus on energy-efficiency."*
- *"Stockholm authorities approve the school renovation project."*

The keywords "school," "renovation," "Stockholm" and "energy-efficiency" are found in the notes. Each unique keyword contributes $C_{\text{Notes}} = 10$ points:

$$4 \times C_{\text{Notes}} = 4 \times 10 = 40 \text{ points.}$$

4.6.3 Similar Words Matches

The similarity score leverages semantic similarity between the user's query and the project's user notes using cosine similarity of their embeddings (described in 4.2). Only notes with a similarity score above a predefined threshold are considered. The sum of these similarity scores is then capped at a maximum value C_{Similar} to prevent it from disproportionately influencing the final ranking. Mathematically:

$$\min \left(\sum_{\substack{\text{note} \in \{\text{similar_notes}\} \\ s(\text{note}, \text{query}) \geq \text{threshold}}} s(\text{note}, \text{query}), C_{\text{Similar}} \right) \quad (4.3)$$

where:

- $s(\text{note}, \text{query})$ is the cosine similarity between the note's embedding and the user's query embedding.
- threshold is the minimum cosine similarity score required for a note to be considered (e.g., 0.6).
- C_{Similar} is the maximum total value (e.g., 5) that the semantic similarity can contribute. Once the sum of note similarities exceeds C_{Similar} , it is capped.

Thus, a project with multiple semantically relevant notes can accumulate similarity points up to a fixed maximum, provided each contributing note meets the similarity threshold.

Example: For the same query, suppose the threshold is set to 0.6 and $C_{\text{Similar}} = 5$ and the project has the following associated user notes:

No.	Note	s(note,query)
1	<i>Renovation plans for the school focus on energy efficiency.</i>	0.75
2	<i>Engaging with local authorities in Stockholm for project approval.</i>	0.65
3	<i>Initial budgeting and resource allocation completed.</i>	0.55
4	<i>Community outreach programs planned post-renovation.</i>	0.60

Table 4.1: Project Notes and Similarity Scores

Notes with a similarity score below the threshold (e.g., Note 3) are excluded. The similarity scores of the remaining notes are summed:

$$0.75 + 0.65 + 0.60 = 2.00 \text{ points.}$$

Since this total does not exceed C_{Similar} , the final similarity score remains 2.00. If the total had exceeded C_{Similar} , it would have been capped at 5. Therefore, in this case, the similarity component contributes 2.00 points to the project's overall score.

4.6.4 Final Score

The final ranking score for each project is the sum of these three components:

$$\text{score} = \underbrace{\sum (\text{Attribute matches}) \times C_{\text{Attr}}}_{(1)} + \underbrace{\sum (\text{Note keywords}) \times C_{\text{Notes}}}_{(2)} + \underbrace{\min(\text{Similarity Score}, C_{\text{Similar}})}_{(3)}$$

The final score of the project "Södermalm International School" with regards to the query "Energy-efficient school renovation in Stockholm" is therefore:

$$45 + 40 + 2 = 87 \text{ points.}$$

All returned projects are ranked in descending order by this final score.

4.7 Generating Results

Structured interviews were chosen to gather focused qualitative insights addressing both the engineering goal—designing a data-driven tool for automated reference project identification—and the scientific inquiry into applying AI for unstructured data in the AEC sector. As recommended by "How to Conduct an Effective Interview; A Guide to Interview Design in Research Study" [42], a structured format ensures consistent questioning and facilitates reliable comparisons across participants. Moreover, alternative quantitative methods were deemed unsuitable for this project because the research questions required in-depth exploration of user experiences, workflows, and data challenges, which necessitated a qualitative approach.

Three key individuals at Cedervall Arkitekter were interviewed: Antonio (BIM Leader), Mia (Tendering Leader), and Ella (Marketing Assistant). They were selected due to their direct involvement in the tendering process, thus providing a broad perspective on technical data management and organizational workflows. Each participant was asked 18 questions divided into three main sections—(1) Introduction & Current Process, (2) Experience & Evaluation of the AI-Driven Tool, and (3) Perceived Impact & Future Outlook. All interviews were conducted in person at Cedervall Arkitekter's Stockholm office, lasting approximately 30 minutes each. The standardized protocol included open-ended questions to prompt detailed responses, supplemented by follow-up questions for further clarification where necessary. All sessions were recorded with consent and transcribed verbatim.

Ultimately, this structured interview process not only shed light on how an AI-based retrieval system could address current inefficiencies and knowledge gaps within Cedervall Arkitekter, but also highlighted broader industry-wide applications for such solutions.

4.8 Summary

The solution hinges on a combination of carefully chosen NLP techniques and lightweight infrastructure, tailored to Cedervall Arkitekter's needs. A multilingual Sentence-BERT variant (`distiluse-base-multilingual-cased-v2`) provides efficient text embeddings for semantic similarity, and the `Helsinki-NLP/opus-mt-en-sv` model handles English-to-Swedish translation. Projects are stored in an on-premise server, ensuring both data governance and reduced operating costs compared to cloud-based alternatives. User interaction occurs through a Python Flask web form, balancing simplicity and reliability, and a custom ranking algorithm further refines retrieval results by integrating keyword matches in specific database fields, user notes, and capped semantic similarity scores. As illustrated in Figure 4.1, the search algorithm takes the user's query—translating it if necessary—and produces a ranked list of relevant project references. This system was validated through discussions and interviews with key staff, confirming its potential to streamline reference retrieval and deliver a practical AI-driven solution for the architectural domain.

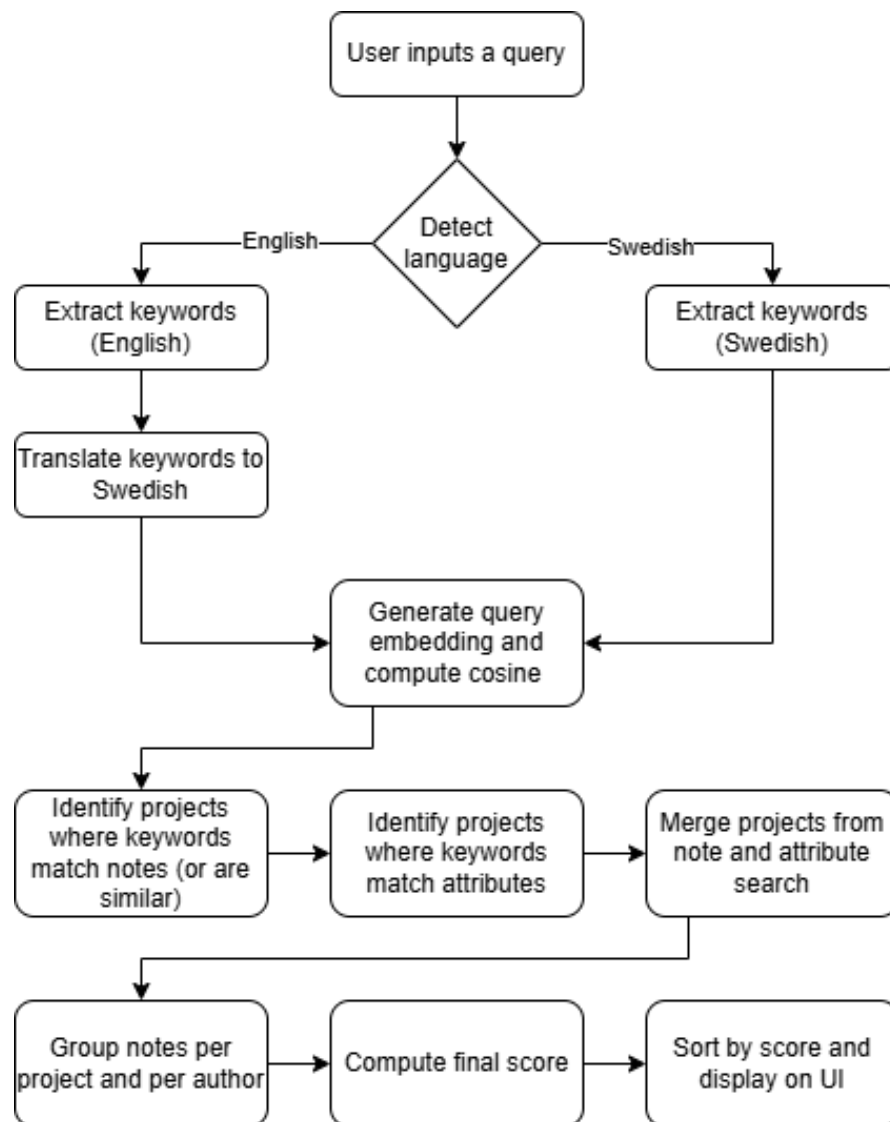


Figure 4.1: Flowchart illustrating the search algorithm.

4.9 Delimitations

This project is specifically confined to utilizing data from Cedervall’s internal systems, particularly the data available through the Milltime system provided by Milient. The solution developed is tailored to the unique data structure and content of Cedervall’s dataset, which includes project metadata and detailed notes or comments entered by employees for each billed hour. Consequently, the quality of the results is directly influenced by the quality of the underlying

data.

The applicability of the system is limited to Cedervall’s data and is not designed as a generic plugin that can be readily implemented with other companies’ data, unless their data conforms precisely to the same structure and format.

The data used is constrained to the fields outlined in 3.1, along with the unstructured textual information from the time entry comments. While these sources provide valuable insights, they may not encompass all aspects that could enhance the retrieval of relevant reference projects. Notably, the project’s scope excludes the integration of Cedervall’s internal file system, which contains a wealth of additional information such as architectural drawings, client interviews, and other detailed documentation.

Incorporating the internal file system presents significant challenges due to its unstructured nature and the diversity of file types it contains, including PDFs, Word documents, images, and other formats. Additionally, the amount and type of data available vary significantly from project to project, making it difficult to standardize and structure the data for analysis. Processing and extracting meaningful information from such a heterogeneous and unstructured dataset would require advanced techniques and resources beyond the scope of this project.

Furthermore, while advancements in cloud services and AI technologies are making strides in natural language processing and data retrieval, there is currently no widely available solution capable of effectively searching a large, unstructured file system for relevant files based on natural language queries in a manner that meets the needs of this project. Existing solutions, such as Microsoft’s OneDrive integration with Copilot, offer limited functionality. For instance, Microsoft 365 Copilot can extract information from multiple files and summarize content but is constrained to processing a small number of files and lacks the capacity to handle large-scale, diverse datasets [43]. Therefore, integrating such technologies is not feasible within the scope of this thesis.

By outlining these boundaries, the project maintains a focused approach, concentrating on developing a functional prototype that operates within the specified parameters. This ensures that efforts are directed toward demonstrating the feasibility and effectiveness of using AI to retrieve relevant reference projects from Cedervall’s existing structured data, without the complexities introduced by broader applicability or additional unstructured data sources.

Chapter 5

Development

The following section provides a concise overview of the development of the reference project retrieval system, illustrating how Cedervall Arkitekter's tender-related challenges were transformed into an automated, on-premise solution. It begins by outlining the initial problem definition and data discovery activities, then describes the transition from large language models to lightweight, specialized NLP approaches. Finally, it covers the system's refinement, including incorporating user notes from the ODBC server and implementing a custom ranking framework. The narrative highlights both the technical decisions made and the iterative feedback from key stakeholders, culminating in a system that effectively balances performance, accuracy, and data security requirements.

5.1 Problem Analysis

Cedervall Arkitekter's procurement team approached this project with a clear but challenging need: to streamline the identification of suitable reference projects in tenders. Their existing process relied heavily on employees' personal memory and a narrow set of "go-to" projects. If new references were ever required—perhaps due to a tender's strict technical criteria or limitations on project age—the procurement team spent several days manually combing through scattered data in hopes of finding a relevant match. This challenge was further compounded by the lack of a robust filtering function within their system. Often, they discovered that the chosen project did not fully meet the criteria, resulting in lost bids. Cedervall saw this as a seemingly impossible situation, with their data stored in various locations and formats, and they had not yet discovered a suitable software solution that could leverage this frag-

mented information.

5.2 Data Exploration

During initial onboarding, Cedervall’s procurement team provided access to Milltime, Silverbucket, and their internal file server to assess which data sources might support automated reference project identification. Silverbucket captured company-wide architectural skills and resource allocation details, while the file server contained a wealth of unstructured documentation, including drawings, interviews, and engineering documents. However, the file server lacked structure, with inconsistent naming conventions and a mix of file formats spread across disparate directories. As a result, the file server proved too unwieldy for this project. Silverbucket’s data, although useful for broader staffing decisions, was deemed less critical for identifying comparable past projects. Milltime, on the other hand, provided a rich repository of project metadata and user-entered notes, which directly linked to each hour billed. This structure—already described in Section 3.1—offered enough valuable information to build a functional retrieval system.

In order to get acquainted with the content and structure of Milltime data, initial experiments were run against a test server that contained older projects (latest from 2018). This provided a safe environment to develop and refine data retrieval logic without compromising ongoing projects. The data was stored locally in an SQLite database, chosen for its reliability and speed given the volume of records involved.

5.3 First Prototypes

Having identified the most promising data source, the next step was to prototype a query interface capable of meeting Cedervall’s requirements. Early tests employed OpenAI’s GPT-4 model, which was able to handle complex questions, such as “Show me all school projects in Stockholm larger than 5,000 square meters,” with high accuracy. However, this approach quickly incurred significant usage fees and took up to a minute to produce results. The recurring cost of tokens, combined with the risk of dependence on a third-party service, made OpenAI’s API infeasible for long-term corporate usage.

An alternative was to explore large language models from Hugging Face, notably Llama-3, to translate natural language queries into SQL on the fly. In many prototypes, Llama-3’s performance was impressive, delivering accurate

SQL queries for Milltime’s structure. Yet its substantial hardware requirements proved problematic. Hosting the model on cloud-based services like Modal provided an interim solution, but the 20-second average latency was higher than desired for real-time usage. Moreover, a subsequent meeting with Cedervall’s team underscored the importance of keeping all data processing within their secure network, rendering external cloud solutions unjustified for sensitive project information.

Faced with both financial and logistical constraints, the development emphasis turned to achieving lightweight performance while preserving functionality. This shift required discarding large language models for direct SQL translation and refocusing on more efficient, smaller-scale NLP components. Meanwhile, a secure on-site environment was prepared so the entire system, including inference for the language processing models, could run internally.

5.4 Integration of User Notes

Towards the end of the development process, access was granted to the company’s ODBC server, which provided employee user notes that were not available through the standard Milltime API. These notes consisted of free-text fields recorded for each billed hour, capturing highly specific or nuanced details about a project’s progress. As discussed in Section 3.1, they offer valuable context beyond standard metadata. Incorporating them became essential for effective project matching.

With large language models off the table, the focus shifted to specialized NLP methods that could handle the linguistic variety in these notes. Simple keyword searches proved insufficient, as architects frequently used varying terminology, primarily in Swedish but occasionally in English. Embedding models, by contrast, transform text into numerical vectors that encapsulate the underlying meaning rather than just surface keywords. This approach enables the system to compare user queries against the notes—even if they do not share identical words—by measuring the proximity of their vector representations in a semantic space. As a result, projects with conceptually similar notes can be identified more reliably, improving the accuracy and relevance of the overall retrieval.

5.5 Architecture and Ranking Logic

The architecture for query processing was designed in a modular fashion. If the submitted query was in English, it was first translated to Swedish using a `MarianMT` model, ensuring all incoming queries were consistently in Swedish for matching. A lightweight language detection tool, `langdetect` [44], determined if translation was required. `spaCy` [45] was used to tokenize the text and remove standard stopwords. The resulting keywords helped filter out obviously irrelevant projects.

To capture deeper semantic meaning, a `SentenceTransformer` model called `distiluse-base-multilingual-cased-v2` was employed. It translated the user query and relevant user notes into embeddings, comparing them via cosine similarity. This embedding-based approach helped uncover projects with notes that might not share exact keywords but had closely related content. After extensive experimentation with multiple thresholds, a similarity cutoff at 0.60 was found to yield robust retrieval performance.

Once prospective matches were identified, they were combined into a single data structure. The final ranking of these projects was governed by a custom scoring system reflecting Cedervall's input. Fifteen points were assigned for each match in the structured attributes, ten points for each unique keyword found in notes, and a maximum of five points for projects whose notes demonstrated strong semantic similarity to the query. These weights were tuned collaboratively with the procurement team, and further details are documented in Section 4.6.

5.6 Deployment, Testing, and Feedback

Upon completion of the core features, direct access to the live Milltime database was enabled. This provided an up-to-date project dataset and necessitated hosting the system on a virtual machine within Cedervall's secure environment. The new hardware substantially improved query response times, allowing searches to complete in roughly one second. Embedding calculations, although more intensive, still ran within a few seconds, meeting the practical expectations for an internal corporate tool.

With the platform running entirely on-premise, data security concerns were minimized. Additional protective measures included a basic login portal restricted to key personnel, SSL certificates to guard against potential man-in-the-middle attacks, and network access limited to employees connected via

the company VPN. This setup aligned with Cedervall's privacy requirements, ensuring no project data traveled outside their internal infrastructure.

To confirm the system's usability, unmoderated user testing was organized for a small group of procurement and technical staff. After a designated trial period, these participants convened for a feedback session. They appreciated the significant reduction in manual search time, although a few implementation bugs surfaced. One issue involved a mismatch between how the system extracted project names and assigned points for keyword matches. Additionally, certain queries produced projects that scored zero due to a missing link between the search function and the ranking mechanism. Once corrected, the system performed as expected. Based on user comments, an information button was added to guide new users through the interface.

These updates concluded the primary development phase. The resulting software is sufficiently modular to allow further enhancements—such as integrating new language models or additional data sources—without disrupting the core architecture. The final product is a robust, on-premise system that significantly cuts down on the time and complexity involved in finding suitable references for tender submissions.

Chapter 6

Results

This section presents the core outcomes of the work undertaken in this thesis. First, an overview of the final product is provided, including screenshots and brief explanations of the user interface developed for optimizing reference project retrieval. Next, key findings from interviews with Cedervall Arkitekter’s procurement team are examined, highlighting how the solution contributes to tangible time savings and enhanced accuracy. Lastly, the chapter illustrates how the methods introduced in this thesis can be extended beyond a single firm’s use case, thereby addressing the broader scientific question of effectively leveraging unstructured internal data for knowledge-intensive tasks in the AEC industry.

6.1 Final Product

Figures 6.1, 6.2 and 6.3 show the user interface of the final software prototype. Users are presented with a search bar where they can enter free-text queries, after which the system retrieves relevant project entries by matching attribute fields and user notes (Swedish: *Anteckningar*), as well as identifying semantically similar text. In line with the approach described in Section 4.6, each project is assigned a score (Swedish: *Poäng*), offering a rapid way to grasp the project’s potential relevance. The interface also includes an edit button to modify automatically extracted keywords, ensuring flexibility when the extraction logic requires refinement. To enhance user-friendliness, an information button is available, which provides opens a window with a quick guide on how the program works. Some columns may display *Unknown* or remain empty if the original data sources lack the relevant information. All design elements align with Cedervall Arkitekter’s existing branding, employing consistent col-

ors and typography. Once a user identifies potentially useful projects, they can reference the project numbers to locate additional documentation on the internal file server, or contact the individuals listed as project managers or note authors for further details.

C&DERVALL

PROJEKTSÖKNING ⓘ

school with a playground in Stockholm ↗

Nyckelord: skola, lekplats, stockholm ↗

Projekt nr	Namn	Anteckningar	Projektledare	Skapad	Sist uppdaterad	Disciplin	Team	Storlek (kvadrat)	Plats	Projektyp	Färdigt	Ny/Ömbyggnad	Art. pr.år	Poäng
42941500	Eglskolan	Frederik Åhlen Möta Eglskolan samt platsbesök. Fäst inredning hus H, hemtutorlärosalar, sarskola, föräld, personal och allrum mm. Molly Miller arbetsmöte med verksamheten på Eglskolan enhetlig kulturenhetens principer i Färdigöring sarskolan och funktionsomläggning hus H och Hus G	Andreas Lund	2018-12-19	2022-03-15	Arkitekt	Team 3	9000	Skärholm en, Stockholm , SWE	Skola	FUBH	Nybyggnad	7700000	70
42944500	St Erik Oskars Skola	Carin Andersson Möta skolan med skolan Skolans skolor Möta LA, genomgång av projekt och bakgrund möten med skola och skola Beatrice Ek Bakal skoleffekt, stockholm stad platsanpassning	Johan Öland	2020-06-09	2022-01-13	Landskap	Team L	7500	Stockholm	Skola	BH	Om- och tillbyggnad	685000	68
35601300	Gårdsbyggnaden	Anna Åberg Stockholm konst möte på skolan med konstnär Möte med Stockholm Konst Stockholm konst möte på skolan med konstnär Möte med Stockholm Konst Micael Larsson tillsammans jämför besöksplan Eglskolan, bokar möte med Skolans ledare	Anna Åberg	2017-06-30	2022-11-01	Arkitekt	Team 6	1500	Årsjåk Stockholm	Skola	FU	Nybyggnad	1240000	67

Figure 6.1: Screenshot of the final user interface in full width. Attribute columns and user notes (*Anteckningar*) appear alongside a relevance score (*Poäng*).

C&D ERVALL
PROJEKTSÖKNING ⓘ

school with a playground in Stockholm **SÖK**

Nyckelord: skola, lekplats, stockholm 🔗

Projekt nr	Namn	Anteckningar	Projektledare
42941500	Egilsskolan	Fredrik Åhlén Möte i Egilsskolan samt platsbesök. Fast inredning hus H, hemvister, läroosalar, särskola, förråd, personal och allrum mm. Molly Miller arbetsmöte med verksamheten på Egilsskolan utredning ljudabsorbenter principer, Fördjupning särskolan och funktionsmöblering hus H och Hus G	Andreas Lund
42944500	St. Erik Oskars Skola	Carin Andersson Möte skyltar med skolan Skolans skyltar Möte LA, genomgång av projekt och bokning av möten med skola och sisab Beatrice Ek Biokol skelettjord, stockholms stad platsanpassning	Johan Öland
35601300	Gårdbyskolan	Anna Årberg Stockholm konst möte på skolan med konstnär Möte med Stockholm Konst Stockholm konst möte på skolan med konstnär Möte med Stockholm Konst	Anna Årberg

Figure 6.2: Screenshot displaying the query and corresponding user notes. The extracted keywords from the query—*skola*, *lekplats*, *stockholm*—are present within the user notes.

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Nyckelord: skola, lekplats, stockholm ✎

Projekt nr
42941500

Namn
Egilsskolan

Anteckningar

Fredrik Åhlén
Möte i Egilsskolan samt platsbesök.
Fast inredning hus H, hemvister, lärosalar, särskola, förråd, personal och allrum mm.

Molly Miller
arbetsmöte med verksamheten på Egilsskolan
utredning ljudabsorbenter principer, Fördjupning särskolan och funktionsmöblering hus H och Hus G

Projektledare
Andreas Lund

Skapad
2018-12-19

Senast ändrad
2022-03-15

Disciplin
Arkitekt

Figure 6.3: Screenshot of the final product displayed in the mobile interface.

6.2 Key Findings from Interviews

To gather reliable qualitative data for the software development effort in this study, three structured interviews were conducted with Cedervall Arkitekter’s procurement team. The methodology for conducting these interviews is detailed in Section 4.7. The participants—Mia, Ella, and Antonio—are responsible for overseeing the reference project selection process during tender submissions. Their roles and experiences provide invaluable insights into the challenges and opportunities within the current workflow.

The key findings from the interviews are grouped into thematic categories: (1) Evaluation of the Data-Driven Tool, (2) Comparison to the Current Procurement Process, and (3) Broader Implications of the Methods.

6.2.1 Participant Overview

The participants oversee the reference project selection process during tender submissions and are expected to be the primary users of the tool. Their roles in Cedervall Arkitekter’s procurement process provide diverse perspectives, which are summarized below.

Participant	Role	Procurement Experience
Antonio	BIM Leader	4 years
Mia	Tendering Leader	1.5 years
Ella	Marketing Assistant	3 years

Table 6.1: Participant Overview

Antonio Heimonen (BIM Leader)

Antonio focuses on tenders related to Building Information Modeling (BIM), coordinating with architects and engineers to meet technical requirements. His responsibilities include managing BIM-specific projects and ensuring that the company’s BIM capabilities are accurately represented in tender submissions.

Mia Modig (Tendering Leader)

Mia oversees the overall tendering process, including identifying potential tenders, coordinating with internal teams, and ensuring submissions meet client requirements. Her role requires significant multitasking and collaboration with other team members.

Ella Sonesson (Marketing Assistant)

Ella plays a critical role in compiling reference projects and ensuring the company's portfolio is accurately represented in tenders. Her work involves extensive research and coordination with project managers and architects to gather relevant project data. Due to her comprehensive knowledge of the company's projects, she serves as a key resource for procurement-related inquiries.

6.2.2 Evaluation of the Data-Driven Tool

This subsection focuses on the interviewees' experiences with the data-driven tool and their assessment of its performance. The participants' feedback provides valuable insights into the tool's effectiveness and its potential to enhance the procurement workflow.

First Impressions and Usability

The interviewees' initial impressions of the AI-driven tool were overwhelmingly positive, with particular emphasis on its ease of use, intuitiveness, and graphical interface. Mia and Antonio described the tool as having a visually appealing design, with Mia stating, *"I'm very happy with the graphics of it and that it's easy to find the information and to understand where the information is coming from."* Similarly, Ella found the tool to be user-friendly and intuitive, noting, *"The data points encapsulate the scope of our requirements in the procurement well."* Ella also highlighted how the tool helped her find small, specific data points in minutes, a task that would normally take hours using the current method.

According to the interviews, the tool's design aligns closely with the users' needs and expectations. Its intuitive interface allows users to quickly input queries and receive relevant results, reducing the learning curve for new users and non-specialists. Notably, the ability to modify keywords and refine searches was particularly appreciated, as it provides flexibility in tailoring the search to specific requirements. Antonio emphasized this feature, stating, *"I'm happy to see that it's possible to modify the keywords if the query isn't correct. That's very useful."* This adaptability ensures that the tool can be customized to meet diverse procurement needs.

Accuracy and Relevance of Recommendations

The interviewees provided valuable insights into the tool's ability to identify relevant reference projects. Overall, the tool was praised for its accuracy and

relevance in matching project requirements. Antonio noted that the tool not only matched his expectations but also surpassed them by uncovering projects he was unaware of, stating, *“It has surprised me how many projects could actually be relevant in tender processes that I didn’t know about.”* This demonstrates the tool’s ability to expand the pool of potential reference projects beyond what is typically considered in the manual process. Additionally, when users knew specific details about the projects they were searching for, the tool consistently delivered accurate results, further reinforcing its reliability.

The tool’s handling of data inaccuracies was also discussed. While the tool relies on data input from Milltime, which can sometimes be incomplete or inconsistent, the interviewees acknowledged that this is not a limitation of the tool itself but rather a reflection of the underlying data. As a result, manual verification remains necessary to ensure the accuracy of the tool’s recommendations.

The tool also bridges knowledge gaps, particularly for newer employees like Mia, who joined Cedervall Arkitekter relatively recently. With only a year and a half of experience in tendering, Mia lacked comprehensive insight into the company’s projects over the past five to ten years. The tool addresses this challenge by providing a centralized and accessible repository of project data, enabling employees with less experience to contribute effectively to the procurement process. This not only enhances confidence in the tool’s output but also supports professional development within the organization.

6.2.3 Comparison to Current Procurement Process

This subsection compares the data-driven tool to Cedervall Arkitekter’s existing procurement workflow, focusing on efficiency, accuracy, and time savings.

Time Savings

The tool significantly reduces the time spent on identifying and compiling reference projects. Mia estimated that the tool could save between 50% and 60% of the time currently spent on this task. Ella similarly noted that the tool reduces waiting times by more than 50%, particularly by eliminating the need for extensive manual searches and coordination with colleagues. For example, as mentioned, Ella highlighted how the tool allowed her to find small, critical data points in just a few minutes, compared to the half-day it would typically take using the current method. Antonio added that the tool reduces the time spent on manual searches and cross-referencing, which are time-consuming aspects of the current workflow.

The tool also reduces dependency on manual searches and meetings. Mia explained that the current process often involves compiling emails, waiting for replies, and coordinating with colleagues, which can be time-consuming. In contrast, the tool streamlines this process, allowing users to find relevant information in minutes rather than hours, greatly reducing the number of meetings and emails. This efficiency enables the procurement team to focus on higher-value tasks, such as refining tender submissions or exploring new opportunities.

Accuracy and Completeness

The tool outperforms the current method in terms of accuracy and completeness when identifying relevant projects. Mia noted that the tool finds a broader range of projects compared to the manual approach, which often relies on reusing the same projects due to familiarity. Antonio also observed that the tool expands the pool of potential references, helping users discover projects they might not have considered otherwise.

A significant issue with the current system is its heavy reliance on the memory of a few key employees, particularly Ella, who described the process as “*pure memory*.” This approach is not only inefficient but also highly unreliable, as it places the burden of recalling project details on a single individual. Ella elaborated on this, stating, “*I’m the person here who has the broadest knowledge of projects due to my role, but that also means I have to know all the projects, and nobody can do that.*” This reliance on memory creates a bottleneck in the procurement process, as Ella’s knowledge is limited to the projects she has personally worked on or been informed about. For example, her year-long parental leave resulted in a gap in her knowledge, leaving the team without a comprehensive understanding of projects completed during that time.

The tool addresses this challenge by providing a centralized and searchable repository of project data, reducing the risk of missing relevant projects due to human error or forgetfulness.

Impact on Workflow and Collaboration

The tool has a transformative impact on workflow and collaboration within the procurement process. It reduces the need for meetings, emails, and phone calls, as highlighted by Mia, who stated that the tool allows her to “*make three phone calls instead of ten.*” Ella also emphasized how the tool cuts down waiting times for responses from colleagues, enabling faster decision-making.

Additionally, the tool empowers non-specialists to contribute more effectively to the procurement process. Mia explained that the tool allows employees who are not primarily involved in tendering to “*get much further in the process*,” enabling the company to distribute tendering work more evenly. This not only improves efficiency but also supports personal and professional development. Mia noted that the tool helps “*offer different people for different jobs*,” allowing employees to take on new roles and develop their skills. This flexibility is particularly valuable in a dynamic work environment, where employees’ availability and expertise can vary.

6.3 Broader Implications of the Methods

The methods developed in this thesis—namely automated data extraction, unstructured text cleaning, embedding-based semantic retrieval, and domain-specific ranking—were applied primarily to improve reference project identification within Cedervall Arkitekter. However, the foundational concepts can be generalized or extended to a broader scope in the AEC industry, where large volumes of heterogeneous, unstructured data are often overlooked or difficult to navigate [46]. This section highlights how the techniques introduced here could address industry-wide challenges and discusses potential next steps for further leveraging AI in knowledge-intensive AEC workflows.

6.3.1 Implications for Architecture Firms

This study primarily focused on optimizing reference project retrieval through data-driven methods; however, these techniques hold potential for optimizing numerous other processes within architecture firms. Although Cedervall Arkitekter served as the principal case study, many architectural offices face similar organizational structures and data challenges. For instance, firms often maintain extensive repositories of unstructured text, including project notes, meeting minutes, and email threads, which frequently remain isolated or disorganized [46]. The embedding-based semantic retrieval pipeline developed in this study can unify these fragmented data sources into an accessible and coherent knowledge base, streamlining tasks that extend well beyond tender preparation, such as internal knowledge management, post-project evaluations, and automated document classification.

The implications for Cedervall Arkitekter specifically are further explored in Section 7.2.

6.3.2 Implications for the AEC Industry at Large

While architecture-focused data served as the foundation of this research, the underlying methods of data preprocessing, sentence embedding, and similarity-based ranking are broadly applicable throughout the AEC domain. Clear parallels can be drawn between this study and other applications across the industry, particularly in construction and civil engineering. These fields also generate vast amounts of unstructured and semi-structured data, including daily site logs, inspection reports, safety records, email communications, and various IoT (Internet of Things) feeds [47, 48]. In many instances, this information is stored in fragmented repositories that limit its potential for collective learning and process optimization. Bilal et al. [47] illustrate how big data strategies can integrate such disparate sources, a principle closely mirrored in the embedding and ranking techniques described here. By adopting semantic retrieval approaches similar to those used for reference project identification in this thesis, AEC teams can quickly find critical documentation or historic performance metrics, therefore promoting more informed decision-making in complex, large-scale projects.

An additional consideration involves early and effective mitigation of risks, which is particularly crucial in large infrastructure projects [48, 49]. While the main focus of this thesis has been on improving reference identification, the same AI pipeline can surface past incidents, best practices, or lessons learned from comparable initiatives, thereby reducing uncertainty in future projects. As discussed by Zabala-Vargas et al. [49], data-driven strategies are increasingly recognized for their potential to anticipate and manage risk factors, especially when organizations systematically gather text-based observations into a unified retrieval system. Embedding-based methods not only link similar incidents across projects but also highlight underlying causes or conditions that could endanger safety, cost, or scheduling outcomes.

A further benefit lies in the ability to query multilingual data, as shown in Section 4.2, which extends to globally dispersed AEC collaborations involving multiple languages and domain terminologies. By standardizing project documentation into vector embeddings, cross-lingual ambiguities can be significantly reduced, enabling geographically distributed teams to share their expertise more fluidly. Multidisciplinary AEC environments often suffer from isolated information flows, but semantic retrieval can bridge these divides by drawing connections among documentation authored by various stakeholders. This not only advances cross-functional learning but also streamlines public procurement tasks that require extensive documentation of certifications and

reference projects. The unification of large-scale unstructured data into a coherent search framework therefore stands to benefit procurement outcomes, operational efficiency, and risk management efforts across the broader AEC sector.

6.3.3 Enhancing Data Quality and Adaptive Retrieval Systems

One key insight from this thesis is that high-quality input data is essential for reliable and meaningful retrieval. To improve this, architecture and engineering offices must implement processes that encourage consistent tagging, note-taking, and documentation. For instance, establishing clear guidelines on how employees enter project notes in systems like Milltime would minimize noise and improve the performance of embedding-based algorithms. Additionally, a simple feedback loop—allowing users to mark retrieved projects as relevant or irrelevant—could incrementally refine future search results in a semi-supervised manner. This would be an interesting topic to explore for future development of the software created in this project.

Although advanced AI models like large language models (LLMs) can further enrich text understanding, this thesis demonstrates that even smaller, well-tuned sentence embedding approaches can yield robust and interpretable results, particularly when combined with domain-specific ranking as discussed in Section 4.6. Thus, organizations with limited computational budgets can still achieve significant gains without fully adopting the most resource-intensive models.

Chapter 7

Discussion

This chapter synthesizes the findings of this thesis, contextualizing them within the broader scientific and engineering challenges outlined in Section 1.2.1. The discussion is divided into three main parts: (1) an analysis of the scientific problem and its implications, (2) an evaluation of the engineering problem and the developed solution, and (3) an exploration of future work that builds on the results of this study. By addressing these aspects, this chapter aims to provide a comprehensive understanding of the contributions made by this thesis and their relevance to both Cedervall Arkitekter and the wider AEC industry.

7.1 Discussion of the Scientific Problem

The scientific problem addressed in this thesis revolves around the challenge of leveraging AI techniques to process and utilize unstructured internal data within the AEC industry, with a focus on optimizing reference project identification in architecture firms. This problem is representative of a broader issue in the AEC sector: the underutilization of vast amounts of unstructured data generated during project lifecycles. While the industry has made strides in adopting digital tools, the effective extraction and application of knowledge from unstructured data remain underexplored.

The results of this study demonstrate that AI-driven methods, such as embedding-based semantic retrieval and domain-specific ranking, can significantly enhance the accessibility and utility of unstructured data. By transforming fragmented and heterogeneous data into a coherent and searchable knowledge base, the developed solution addresses a critical bottleneck in Cedervall Arkitekter's tendering process. This achievement not only validates the feasibility of applying AI techniques to unstructured data in the AEC indus-

try but also highlights the potential for these methods to be extended to other knowledge-intensive tasks, such as risk analysis, best practice identification, and resource allocation [48, 49].

Furthermore, the study underscores the importance of domain-specific adaptations in AI applications. While generic AI models can provide a foundation, their effectiveness is greatly enhanced when tailored to the unique characteristics of the AEC industry. For instance, the integration of domain-specific ranking criteria, as discussed in Section 4.6, ensures that the retrieved results align with the practical needs of procurement teams. This finding has broader implications for the AEC industry, suggesting that future AI implementations should prioritize domain-specific customization to maximize their impact.

7.2 Discussion of the Engineering Problem

The engineering problem tackled in this thesis involved designing and implementing a data-driven solution to automate the identification of relevant reference projects from Cedervall Arkitekter's vast unstructured internal datasets. This problem was motivated by the inefficiencies and resource-intensive nature of the firm's existing procurement process, particularly the manual identification of reference projects. The developed solution successfully addresses this challenge by combining automated data extraction, text cleaning, semantic retrieval, and ranking into a cohesive tool.

The key strength of the solution lies in its ability to streamline the reference project identification process, reducing the time and effort required while improving accuracy and completeness. As evidenced by the interviews with Cedervall Arkitekter's procurement team (Section 6.2.2), the tool not only saves significant time but also uncovers projects that might otherwise be overlooked due to reliance on human memory or incomplete data.

The company has expressed strong satisfaction with the solution, and at the time of writing, it has already been integrated into their daily processes. The procurement team reports that the tool has significantly reduced the time spent on identifying reference projects, with some tasks now taking minutes rather than hours. This immediate impact underscores the practical value of the solution and its alignment with the company's needs. By automating a previously labor-intensive process, the tool has not only enhanced efficiency but also empowered employees to focus on higher-value tasks, further contributing to the firm's operational success.

It can therefore be concluded with confidence that the project has successfully enhanced the efficiency of the tendering process at Cedervall Arkitekter,

effectively solving the identified engineering problem.

7.3 Future Work

This project opens several avenues for future work, encompassing both scientific research and practical development. The following subsections outline potential directions for further exploration and enhancement.

7.3.1 Scientific Research

A promising direction for future research involves extending the methodologies discussed in this thesis to other use cases within the AEC industry. For instance, the application of these methods could be explored in the context of risk and incident analysis, where structured and unstructured data are utilized to identify patterns and mitigate potential issues. Additionally, the approach could be adapted to analyze best practices in large-scale infrastructure projects, thereby contributing to the optimization of project management and execution within the industry.

7.3.2 Engineering and Development

From a development perspective, the project offers numerous opportunities for expansion and refinement. Key areas for improvement include the integration of advanced technologies, user interface enhancements, system integrations, and adaptive retrieval mechanisms.

First, reintroducing a Large Language Model into the system, given the availability of appropriate hardware, would significantly enhance the program's capabilities. While the current implementation delivers satisfactory results, the inclusion of an LLM would introduce a "smart" element, enabling the system to handle highly specific queries and address edge cases that are currently beyond its scope.

Second, several low-effort, high-impact improvements can be made to the user interface. For example, incorporating filtering and sorting functionalities, like those found in Excel, would improve usability. Additionally, an export feature could be implemented to allow users to seamlessly transfer data to Excel for further analysis.

Third, integrating the tool with Silverbucket, a resource management platform, would enable the system to identify suitable employees for projects proposed in tenders. By leveraging the tool's existing knowledge of projects, it

could match employees to projects based on criteria such as skill sets, experience, and availability, thereby optimizing resource allocation.

Fourth, restructuring the file system to facilitate the extraction of additional data would enrich the Milltime dataset, providing a more comprehensive foundation for analysis and decision-making. Furthermore, linking architectural drawings and project-related interviews directly within the tool would streamline access to critical information, ensuring that users are only one click away from relevant resources.

Fifth, implementing an adaptive retrieval mechanism could significantly improve the accuracy and relevance of search results over time. Introducing a feedback loop that allows users to mark retrieved projects as relevant or irrelevant would enable the system to refine future recommendations in a semi-supervised manner. Additionally, enforcing structured tagging and documentation practices within architectural firms, such as clear guidelines for entering project notes in systems like Milltime, would help minimize noise and improve data consistency, further enhancing retrieval performance.

Finally, implementing all these features would transform the tool into a holistic procurement system capable of significantly accelerating the company's processes. While the initial focus of this project was on optimizing the reference project identification phase—identified as the primary bottleneck—future work could expand the system's scope to address additional aspects of the procurement process.

7.4 Reflections

Many lessons emerged from the development of this project. By investigating a range of AI tools and understanding how they can be adapted for specific business challenges, I developed a deeper appreciation for the diversity of advanced algorithms and their potential in solving real-world problems. My coding skills also improved significantly, as the project required careful integration of data retrieval, preprocessing pipelines, and natural language processing within a new domain. Prior to this thesis, I had limited insight into the architect industry; engaging directly with Cedervall Arkitekter broadened my perspective on how technology can be harnessed to support more traditional fields.

A principal obstacle involved navigating organizational constraints within the external company setting. Understanding Cedervall Arkitekter's internal data structures proved difficult, partly because the firm itself lacked comprehensive documentation of how the data was stored or accessed. This challenge

was exacerbated by restricted permissions: initially, I had access only to the test database containing outdated projects that were no longer relevant for real-world tenders. Although developing prototypes on this data helped refine my approach, it did not immediately benefit Cedervall Arkitekter's active procurement needs.

Further complications arose from stringent security measures that delayed my access to the live Milltime database and the ODBC server containing user notes. When these permissions were finally granted, key development tasks—such as integrating real data and establishing the on-premise hosting environment—had to be compressed into the later stages of the project. Limitations on external hosting services meant rethinking certain architectural decisions, particularly regarding resource-heavy tasks that could not be offloaded to cloud-based platforms. Despite these hurdles and time constraints, the final system met Cedervall Arkitekter's requirements, demonstrating the adaptability of the core design.

In a broader sense, this project exemplifies the practical essence of software engineering. At its foundation, software engineering seeks to produce robust, efficient, and maintainable systems that solve specific problems. The solution presented here required the management of multiple interconnected components, spanning database handling, machine learning model integration, security and access controls, and user-focused interface design. Balancing these technical aspects with real-world constraints—such as legacy data repositories, corporate security policies, and limited computational resources—mirrors many of the complexities inherent in professional software projects. The effort to ensure reliability, usability, and stakeholder acceptance reflects the iterative, interdisciplinary nature of software engineering itself.

Overall, delivering a functional product that brought tangible efficiency gains to Cedervall Arkitekter's workflow was highly rewarding. Observing its immediate impact—both in terms of staff adoption and positive feedback—reinforced the importance of clear communication, robust planning, and domain-specific solutions. These experiences will undoubtedly inform my approach to future projects, whether within the AEC sector or other industries that rely on complex data and demand innovative, secure, and scalable software solutions.

Chapter 8

Conclusion

This thesis investigated the application of artificial intelligence techniques to streamline the identification of reference projects within architectural tenders. The existing process at Cedervall Arkitekter relied heavily on manual searches, employee memory, and unstructured data scattered across multiple systems. To address these inefficiencies, an AI-driven tool was designed and implemented, integrating natural language processing, semantic similarity models, and a structured ranking system.

The development process involved several phases, including data exploration, method selection, and system deployment. Initially, experiments were conducted using large-scale natural language models, but due to their computational demands and security constraints, a more lightweight embedding-based retrieval approach was adopted. The final system, deployed on-premise, enables users to enter free-text queries and retrieve ranked project references based on a combination of structured attributes, user note keyword matches, and semantic similarity scores.

Evaluation of the tool included structured interviews and user testing with key stakeholders at Cedervall Arkitekter. The results indicate that the system significantly reduces the time required to find relevant reference projects, improving accuracy and reducing dependency on personal recollections. The ability to modify keywords and refine search criteria was particularly valued by users, who also emphasized the system's potential for broader applications beyond procurement.

From an engineering perspective, this project exemplifies the complexities of integrating AI solutions into real-world corporate environments. Balancing computational efficiency, data security, and user accessibility was essential to ensuring successful adoption. Furthermore, the work highlights the impor-

tance of domain-specific adaptations when applying AI in industry-specific contexts.

In conclusion, this thesis demonstrates that AI-driven retrieval systems offer tangible benefits in architectural tenders by automating knowledge-intensive tasks, improving efficiency, and reducing manual workload. The findings suggest that similar AI methodologies could be applied across the broader AEC industry to enhance data-driven decision-making in procurement and beyond.

Bibliography

- [1] Encyclopædia Britannica. History of artificial intelligence, 2024. Retrieved September 27, 2024, from <https://www.britannica.com/science/history-of-artificial-intelligence>.
- [2] Roberto Cordeschi. Ai turns fifty: Revisiting its origins. In Applied Artificial Intelligence, volume 21, pages 259–279, 4 2007.
- [3] IBM. Early games, 2024. Retrieved September 27, 2024, from <https://www.ibm.com/history/early-games>.
- [4] Luigi De Angelis, Francesco Baglivo, Guglielmo Arzilli, Gaetano Pierpaolo Privitera, Paolo Ferragina, Alberto Eugenio Tozzi, and Caterina Rizzo. Chatgpt and the rise of large language models: the new ai-driven infodemic threat in public health. Frontiers in Public Health, 11, 2023.
- [5] Daya Guo, Qihao Zhu, Dejian Yang, Zhenda Xie, Kai Dong, Wentao Zhang, Guanting Chen, Xiao Bi, Y. Wu, Y. K. Li, Fuli Luo, Yingfei Xiong, and Wenfeng Liang. Deepseek-coder: When the large language model meets programming – the rise of code intelligence, 2024.
- [6] Hamed Nabizadeh Rafsanjani and Amir Hossein Nabizadeh. Towards human-centered artificial intelligence (ai) in architecture, engineering, and construction (aec) industry. Computers in Human Behavior Reports, 11:100319, 8 2023.
- [7] Wrike Blog. Navigating the aec industry: What is aec and how is the industry changing?, 2023. Accessed: 2024-11-01.
- [8] Sofiat O. Abioye, Lukumon O. Oyedele, Lukman Akanbi, Anuoluwapo Ajayi, Juan Manuel Davila Delgado, Muhammad Bilal, Olugbenga O. Akinade, and Ashraf Ahmed. Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges, 12 2021.

- [9] Cedervall Arkitekter. Om oss, 2024. Retrieved September 27, 2024, from <https://cedervallarkitekter.se/om-oss/>.
- [10] Cedervall Arkitekter. Projects, 2024. Retrieved September 26, 2024, from <https://cedervallarkitekter.se/projekt/>.
- [11] Per Erik Eriksson, Leentje Volker, Anna Kadefors, Sofia Lingegård, Johan Larsson, and Lilly Rosander. Collaborative procurement strategies for infrastructure projects: A multiple-case study. Proceedings of Institution of Civil Engineers: Management, Procurement and Law, 172:197–205, 10 2019.
- [12] United Nations. Economic growth, 2024. Retrieved September 27, 2024, from <https://www.un.org/sustainabledevelopment/economic-growth/>.
- [13] United Nations. Infrastructure and industrialization, 2024. Retrieved September 27, 2024, from <https://www.un.org/sustainabledevelopment/infrastructure-industrialization/>.
- [14] Andreas Öjehag Pettersson and Mikael Granberg. Public procurement as marketisation: Impacts on civil servants and public administration in sweden sjpa 23(3-4). Scandinavian Journal of Public Administration (SJPA), 2019.
- [15] Government Offices of Sweden. Public procurement - how it works in sweden, n.d. Retrieved October 8, 2024, from <https://www.government.se/government-policy/central-government-administration/public-procurement---how-it-works-in-sweden/>.
- [16] Upphandlingsmyndigheten (Swedish Public Procurement Authority). About public procurement, n.d. Retrieved October 8, 2024, from <https://www.upphandlingsmyndigheten.se/en/about-publicprocurement/#:~:text=Public%20procurement%20is%20the%20purchasing,contribute%20to%20sustainable%20societal%20development.>
- [17] Amos Darko, Albert P.C. Chan, Michael A. Adabre, David J. Edwards, M. Reza Hosseini, and Ernest E. Ameyaw. Artificial intelligence in the

- aec industry: Scientometric analysis and visualization of research activities. Automation in Construction, 112:103081, 4 2020.
- [18] Adetayo Olugbenga Onososen and Innocent Musonda. Perceived benefits of automation and artificial intelligence in the aec sector: An interpretive structural modeling approach. Frontiers in Built Environment, 8, 4 2022.
 - [19] Michela Guida, Federico Caniato, Antonella Moretto, and Stefano Ronchi. The role of artificial intelligence in the procurement process: State of the art and research agenda. Journal of Purchasing and Supply Management, 29, 3 2023.
 - [20] Lucia Siciliani, Vincenzo Taccardi, Pierpaolo Basile, Marco Di Ciano, and Pasquale Lops. Ai-based decision support system for public procurement. Information Systems, 119, 10 2023.
 - [21] Oracle Corporation. Oracle procurement cloud. <https://www.oracle.com/procurement/>, 2021.
 - [22] D. M. Teixeira. Introducing the new oracle digital assistant sql dialog. Oracle Blogs, January 2023.
 - [23] SAP SE. Sap ariba. <https://www.ariba.com/>, 2021.
 - [24] GEP. Gep smart procurement software. <https://www.gep.com/software/gep-smart>, 2021.
 - [25] Milient AB. Milient project management solutions. <https://www.milient.se/>, 2023. Accessed: Nov. 1, 2024.
 - [26] European Parliament and Council of the European Union. Regulation (eu) 2016/679 (general data protection regulation). Official Journal of the European Union, 2016.
 - [27] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In Proc. of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), pages 4171–4186, 2019.
 - [28] Y. Liu et al. Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692, 2019.

- [29] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing, pages 3982–3992, 2019.
- [30] Jörg Tiedemann and Santhosh Thottingal. Helsinki-nlp/opus-mt-en-sv. <https://huggingface.co/Helsinki-NLP/opus-mt-en-sv>, 2020. Accessed: 2025-01-15.
- [31] Jörg Tiedemann and Santhosh Thottingal. Opus-mt-building open translation services for the world, 2020.
- [32] F. Xu et al. Sqlcoder: Neural sql code generation. In Proc. of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 1160–1170, 2022.
- [33] OpenAI. Openai api. <https://platform.openai.com>, 2023. Accessed: 2025-01-16.
- [34] H. Touvron et al. Llama 2: Open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288, 2023.
- [35] Defog AI. Llama-3 sqlcoder 8b. <https://huggingface.co/defog/llama-3-sqlcoder-8b>, 2024. Accessed: 2025-01-15.
- [36] Tim Dettmers, Mike Lewis, Sam Shleifer, and Luke Zettlemoyer. Bitsandbytes: 8-bit optimized gpu memory management for transformers. <https://pypi.org/project/bitsandbytes/>, 2022. Accessed: 2025-01-15.
- [37] Modal Labs. Modal: Serverless stack for data teams. <https://modal.com/>, 2023. Accessed: Nov. 1, 2024.
- [38] T. Bocklisch, J. Faust, N. Gruenewald, and A. Nichol. Rasa: Open source language understanding and dialogue management. arXiv preprint arXiv:1712.05181, 2017.
- [39] Botpress Inc. Botpress: The conversational ai platform. <https://botpress.com/>, 2023. Accessed: Nov. 1, 2024.
- [40] Juan Ramos. Using tf-idf to determine word relevance in document queries, 2003.

- [41] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. Foundations and Trends in Information Retrieval, 3:333–389, 2009.
- [42] Hamed Taherdoost. How to conduct an effective interview; a guide to interview design in research study, 2022.
- [43] Microsoft Corporation. Get started with copilot in onedrive. <https://tinyurl.com/3uv266xp>, 2023. Accessed: Nov. 1, 2024.
- [44] Michal Danilák. langdetect - python language detection library. <https://pypi.org/project/langdetect/>, 2024. Accessed: 2024-02-01.
- [45] Explosion AI. spacy models and pipelines. <https://spacy.io/usage/models>, 2024. Accessed: 2024-02-01.
- [46] Hongqin Fan, Fan Xue, and Heng Li. Project-based as-needed information retrieval from unstructured aec documents. Journal of Management in Engineering, 31, 1 2015.
- [47] Muhammad Bilal, Lukumon O. Oyedele, Junaid Qadir, Kamran Munir, Saheed O. Ajayi, Olugbenga O. Akinade, Hakeem A. Owolabi, Hafiz A. Alaka, and Maruf Pasha. Big data in the construction industry: A review of present status, opportunities, and future trends, 8 2016.
- [48] Nitin Liladhar Rane. Integrating leading-edge artificial intelligence (ai), internet of things (iot), and big data technologies for smart and sustainable architecture, engineering and construction (aec) industry: Challenges and future directions. International Journal of Data Science and Big Data Analytics, 3:73–95, 11 2023.
- [49] Sergio Zabala-Vargas, María Jaimes-Quintanilla, and Miguel Hernán Jimenez-Barrera. Big data, data science, and artificial intelligence for project management in the architecture, engineering, and construction industry: A systematic review, 12 2023.

